

Substantive Citation Report

Twenty-two substantive citing articles in two groups: articles in A*-rated conferences or with at least 100 Google Scholar citations, and articles whose methods rely on or extend TQT.

22

selected articles

103

marked discussion pages

30

countries / territories in
the wider survey

Selection and source documentation

Group A includes articles published in conferences rated ICORE2026 A* and/or carrying at least 100 Google Scholar citations in the saved snapshot; they document methodological adaptation, technical exposition, comparison and quantizer-design connections. Group B includes articles whose own methods or experiments use, follow or extend TQT, or compare an implemented extension against it, as stated in the source passages; background mentions and statements that a design is merely similar to TQT are excluded. Counts belong to the citing articles' Scholar records or version clusters; three Group B records show no cited-by count in the saved snapshot. Each entry identifies its reviewed publication version, the citing authors' affiliations, any employer link and the contribution supported by the source passage.

Authorship links and documented scholarly use

Each article identifies the cited work and authors, the specific research connection and the original discussion pages. The bibliography link establishes authorship of the corresponding work by Sambhav R. Jain and coauthors.

Orange rectangles mark selected discussion, citations and TQT results; blue rectangles identify the corresponding bibliography entries. Pages 1–103 of the marked packet follow the article order in this report and retain original page numbers. Article numbers identify the corresponding marked files.

International citation survey. The wider survey maps 210 citing works to 228 institution/location entries across 30 countries/territories. These totals describe the broader citation network, separately from this report's selection.

[Report PDF](#) · [Marked discussions](#) · [pages 1–103](#) · [Interactive map](#)

Snapshot: October 7, 2026. [Sambhav Jain's Google Scholar profile](#). Counts belong to the saved citing-article records or version clusters.

A. Articles in A*-rated venues or with at least 100 citations (10)

No.	Article / venue	Scholar	Selection basis
01	<u>Understanding and Overcoming the Challenges of Efficient Transformer Quantization</u> EMNLP 2021	<u>271</u>	ICORE2026 A* venue + ≥ 100 Scholar citations
02	<u>Up or Down? Adaptive Rounding for Post-Training Quantization</u> ICML 2020	<u>1,224</u>	ICORE2026 A* venue + ≥ 100 Scholar citations
03	<u>A White Paper on Neural Network Quantization</u> arXiv / Qualcomm white paper, 2021	<u>1,362</u>	≥ 100 Scholar citations
04	<u>Mixed Precision DNNs: All you need is a good parametrization</u> ICLR 2020	<u>230</u>	ICORE2026 A* venue + ≥ 100 Scholar citations
05	<u>Bayesian Bits: Unifying Quantization and Pruning</u> NeurIPS 2020	<u>207</u>	ICORE2026 A* venue + ≥ 100 Scholar citations
06	<u>Bringing AI to edge: From deep learning's perspective</u> Neurocomputing 485 (2022)	<u>247</u>	≥ 100 Scholar citations
07	<u>HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs</u> ECCV 2020	<u>95</u>	ICORE2026 A* venue
08	<u>QKD: Quantization-aware Knowledge Distillation</u> arXiv preprint, 2019	<u>125</u>	≥ 100 Scholar citations
14	<u>F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization</u> ICLR 2022	<u>72</u>	ICORE2026 A* venue
15	<u>Improving Low-Precision Network Quantization via Bin Regularization</u> ICCV 2021, 5241-5250	<u>65</u>	ICORE2026 A* venue

B. Articles relying on or extending TQT (12)

No.	Article / venue	Scholar	Selection basis
12	Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy Electronics 10(22), 2823 (2021)	15	Relies on or extends TQT
16	A Reduced-Precision Network for Image Reconstruction ACM Transactions on Graphics 39(6), Article 231 (2020)	41	Relies on or extends TQT
17	Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737-10746 (2023)	12	Relies on or extends TQT
18	FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022)	50	Relies on or extends TQT
19	7 μJ/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks Future Generation Computer Systems 149 (2023), 717-731	12	Relies on or extends TQT
20	Optimizing the Deployment of Tiny Transformers on Low-Power MCUs IEEE Transactions on Computers 74(2), 526-541 (2025)	47	Relies on or extends TQT
21	A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps CVPR Workshops 2022 (CVPRW), 2686-2695	12	Relies on or extends TQT
22	A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition IEEE Access 9 (2021), 104367-104381	54	Relies on or extends TQT
23	Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach Computers and Electrical Engineering 134 (2026), 111101	None shown	Relies on or extends TQT
24	INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171-176	None shown	Relies on or extends TQT
25	FPGA-Based Real-Time Embedded Fish Embryo Detection System Mobile Information Systems, vol. 2022, 8021288	None shown	Relies on or extends TQT
26	Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025)	6	Relies on or extends TQT

A. ARTICLES IN A*-RATED VENUES OR WITH AT LEAST 100 CITATIONS

ARTICLE 01 · SUBSTANTIVE CITATION REPORT

Understanding and Overcoming the Challenges of Efficient Transformer Quantization

Yelysei Bondarenko; Markus Nagel; Tijmen Blankevoort
EMNLP 2021 · 271 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. Qualcomm researchers explicitly adapt the jointly credited LSQ/TQT procedure from Esser et al. and Jain et al. to learn weight and activation ranges for BERT-like transformer models.

Methodological adaptation. Adapts trainable weight and activation ranges to BERT-like models, jointly crediting the procedures of Jain et al. and Esser et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 11](#).

Reviewed version: Published proceedings PDF. **Pinpoint:** Physical PDF pp. 3, 7, learnable weight/activation ranges and their adaptation to transformers; bibliography p. 11 / printed p. 7957. Source E-0046; marked copy E-0884.

Count: E-0430, data.organic_results[6].inline_links.cited_by.total. Venue: E-0326.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 01-Transformer-Quantization-MARKED.pdf

ARTICLE 02 · SUBSTANTIVE CITATION REPORT

Up or Down? Adaptive Rounding for Post-Training Quantization

Markus Nagel; Rana Ali Amjad; Mart van Baalen; Christos Louizos; Tijmen Blankevoort
ICML 2020 · 1,224 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. The AdaRound paper names Jain et al., alongside Esser et al., when explaining how quantization ranges can be learned during training instead of being set manually.

Named technical recognition. Explains the contribution of learning quantization ranges in the related-work analysis of AdaRound, with joint credit to Jain et al. and Esser et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Published PMLR PDF; physical pages retained. **Pinpoint:** Physical PDF p. 5, section 4 Background and related work; bibliography physical p. 10. Source E-0321; marked copy E-0885.

Count: E-0430, data.organic_results[2].inline_links.cited_by.total. Venue: E-0329.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 02-AdaRound-MARKED.pdf

A. ARTICLES IN A*-RATED VENUES OR WITH AT LEAST 100 CITATIONS

ARTICLE 03 · SUBSTANTIVE CITATION REPORT

A White Paper on Neural Network Quantization

Markus Nagel; Marios Fournarakis; Rana Ali Amjad; Yelysei Bondarenko; Mart van Baalen; Tijmen Blankevoort
arXiv / Qualcomm white paper, 2021 · 1,362 Google Scholar citations

≥100 Scholar citations

Documented contribution. The quantization white paper explains learning quantization parameters through the straight-through estimator, credits Jain et al. alongside Esser et al. and Bhargat et al., and develops the corresponding gradient derivation.

Technical exposition. Develops the gradient calculation for learnable quantization parameters after crediting Jain et al., Esser et al. and Bhargat et al.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 26](#).

Reviewed version: Reviewed arXiv white paper. **Pinpoint:** Physical PDF pp. 20, 22, gradient derivation and QAT initialization; bibliography p. 26. Source E-0090; marked copy E-0886.

Count: E-0430, `data.organic_results[1].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 03-White-Paper-MARKED.pdf

ARTICLE 04 · SUBSTANTIVE CITATION REPORT

Mixed Precision DNNs: All you need is a good parametrization

Stefan Uhlich; Lukas Mauch; Fabien Cardinaux; Kazuki Yoshiyama; Javier Alonso García; Stephen Tiedemann; Thomas Kemp; Akira Nakamura
ICLR 2020 · 230 Google Scholar citations

ICORE2026 A* venue + ≥100 Scholar citations

Documented contribution. Sony researchers describe TQT's trained dynamic range and train a TQT-style fixed-bitwidth model for their experimental comparison with mixed-precision quantization.

Implemented experimental comparison. Trains a TQT-style model as a fixed-bitwidth benchmark for the authors' concurrently developed mixed-precision method.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10](#).

Reviewed version: Author-hosted arXiv copy bearing ICLR 2020 publication header. **Pinpoint:** PDF/printed pp. 1–2, 4–5, 8–9, introduction, quantizer constraints, experimental setup and TQT result rows; bibliography p. 10. Source E-0087; marked copy E-0887.

Count: E-0430, `data.organic_results[11].inline_links.cited_by.total`. Venue: E-0328.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 04-Mixed-Precision-DNNs-MARKED.pdf

A. ARTICLES IN A*-RATED VENUES OR WITH AT LEAST 100 CITATIONS

ARTICLE 05 · SUBSTANTIVE CITATION REPORT

Bayesian Bits: Unifying Quantization and Pruning

Mart van Baalen; Christos Louizos; Markus Nagel; Rana Ali Amjad; Ying Wang; Tijmen Blankevoort; Max Welling
NeurIPS 2020 · 207 Google Scholar citations

ICORE2026 A* venue + ≥ 100 Scholar citations

Documented contribution. The Bayesian Bits authors credit two papers with independently introducing joint learning of scale and model parameters. Reference [15] identifies the TQT work by Jain, Gural, Wu and Dick.

Attribution of a research contribution. Credits reference [15] as one of two papers that independently introduced joint learning of quantization scale and network parameters.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 10.](#)

Reviewed version: Published NeurIPS PDF. **Pinpoint:** PDF/printed pp. 6, 8, scale learning and TQT in Figure 2b and its caption; bibliography p. 10, reference [15]. Source E-0324; marked copy E-0888.

Count: E-0430, `data.organic_results[9].inline_links.cited_by.total`. Venue: E-0331.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 05-Bayesian-Bits-MARKED.pdf

ARTICLE 06 · SUBSTANTIVE CITATION REPORT

Bringing AI to edge: From deep learning's perspective

Di Liu; Hao Kong; Xiangzhong Luo; Weichen Liu; Ravi Subramaniam
Neurocomputing 485 (2022) · 247 Google Scholar citations

≥ 100 Scholar citations

Documented contribution. The edge-AI survey gives Jain et al. a dedicated methodological description of trained uniform quantization for accurate and efficient neural-network inference on fixed-point hardware.

Named survey discussion. Explains trained uniform quantization for fixed-point inference in a survey of deep learning at the edge; reference [121] identifies Jain et al.'s paper.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. [Bibliography p. 20.](#)

Reviewed version: arXiv author manuscript; journal publication verified separately. **Pinpoint:** PDF/printed pp. 8–9, TQT comparison-table row and named quantization discussion; bibliography p. 20, reference [121]; affiliations p. 1. Source E-0044; marked copy E-0889.

Count: E-0366, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 06-Bringing-AI-to-Edge-MARKED.pdf

ARTICLE 07 · SUBSTANTIVE CITATION REPORT

HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs

Hai Victor Habi; Roy H. Jennings; Arnon Netzer

ECCV 2020 · 95 Google Scholar citations

ICORE2026 A* venue

Documented contribution. HMQ states that the quantizer it uses is similar to the quantizer of Jain et al. identified by reference [24], then presents its hardware-friendly mixed-precision formulation.

Quantizer-design connection. Identifies the quantizer in reference [24] as similar to the one used in HMQ, followed by signed and unsigned quantizer definitions in equations 2 and 3.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 17](#).

Reviewed version: Author arXiv manuscript; ECCV publication verified separately. **Pinpoint:** Physical PDF pp. 1–4, hardware-friendly quantization citations and quantizer definition; bibliography p. 17. Source E-0043; marked copy E-0891.

Count: E-0368, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-0585.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 07-HMQ-MARKED.pdf

ARTICLE 08 · SUBSTANTIVE CITATION REPORT

QKD: Quantization-aware Knowledge Distillation

Jangho Kim; Yash Bhalgat; Jinwon Lee; Chirag Patel; Nojun Kwak

arXiv preprint, 2019 · 125 Google Scholar citations

≥100 Scholar citations

Documented contribution. QKD describes trainable intervals in LSQ and TQT alongside other trainable quantization approaches, and states that it uses these approaches in its baseline before applying knowledge distillation.

Use in a research implementation. Includes TQT [17] among the jointly credited trainable quantization approaches used in the baseline. The reported results concern quantization combined with knowledge distillation.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9](#).

Reviewed version: Reviewed arXiv manuscript. **Pinpoint:** Physical PDF pp. 2–3, 7, trainable intervals, dequantization and depthwise-layer sensitivity; bibliography p. 9, reference [17]. Source E-0056; marked copy E-0892.

Count: E-0378, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 08-QKD-MARKED.pdf

ARTICLE 14 · SUBSTANTIVE CITATION REPORT

F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization

Qing Jin; Jian Ren; Richard Zhuang; Sumant Hanumante; Zhengang Li; Zhiyu Chen; Yanzhi Wang; Kaiyuan Yang; Sergey Tulyakov

ICLR 2022 · 72 Google Scholar citations

ICORE2026 A* venue

Documented contribution. Snap Inc., Northeastern University and Rice University researchers position F8Net against the power-of-two fixed-point scaling of Jain et al. (2019), state that their work differs from it in three major aspects and report TQT as a comparison method in their MobileNet results.

Comparison and design positioning. Describes Jain et al. as restricting all scaling factors to power-of-two values (pp. 2–3), contrasts F8Net's on-the-fly fractional length with directly trained fractional length (p. 7) and lists TQT (Jain et al., 2019) in Table 1 for MobileNet V1 and V2b (p. 8).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 12](#).

Reviewed version: arXiv:2202.05239 PDF carrying the ICLR 2022 conference header. **Pinpoint:** Physical PDF pp. 2–3, 7–8: positioning against the power-of-two fixed-point scaling of Jain et al. (2019), contrast with directly trained fractional lengths and TQT rows in Table 1; bibliography p. 12. Source E-0725; marked copy E-1397.

Count: E-0381, data.organic_results[1].inline_links.cited_by.total. Venue: E-0328.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 14-F8Net-MARKED.pdf

ARTICLE 15 · SUBSTANTIVE CITATION REPORT

Improving Low-Precision Network Quantization via Bin Regularization

Tiantian Han; Dong Li; Ji Liu; Lu Tian; Yi Shan

ICCV 2021, 5241–5250 · 65 Google Scholar citations

ICORE2026 A* venue

Documented contribution. Xilinx researchers in Beijing re-implement a hardware-friendly quantization baseline following references [19, 21], where [19] is TQT, and apply their bin regularization on top of TQT. Their Table 9 reports TQT, TQT+L2 and TQT+BR results for 4-bit MobileNetV2. Employer-linked: the authors list Xilinx Inc.

Extension of the TQT baseline. Builds bin regularization on the TQT baseline: Table 9 reports 63.8% top-1 for TQT and 67.5% for TQT+BR on 4-bit MobileNetV2, which the text describes as +3.7% over TQT.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9](#).

Reviewed version: IEEE Xplore published print, ICCV 2021 proceedings; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 8 (printed p. 5248): Section 4.8 hardware-friendly baseline following [19, 21], Table 9 TQT, TQT+L2 and TQT+BR rows and the reported +3.7% over TQT; bibliography p. 9 (printed p. 5249), reference [19]. Source E-1437; marked copy E-1398.

Count: E-0391, data.organic_results[7].inline_links.cited_by.total. Venue: E-1396.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 15-Bin-Regularization-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 12 · SUBSTANTIVE CITATION REPORT

Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy

Maarten Vandersteegen; Kristof Van Beeck; Toon Goedemé
Electronics 10(22), 2823 (2021) · 15 Google Scholar citations

Relies on or extends TQT

Documented contribution. KU Leuven researchers explicitly state that they optimize power-of-two scales through backpropagation as proposed by Jain et al. They also discuss TQT's shared quantizer parameters for elementwise addition.

Direct methodological adoption. Implements the power-of-two scale backpropagation proposed by Jain et al. on page 12 and describes shared quantizer parameters for elementwise addition.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 23](#).

Reviewed version: Published journal PDF from KU Leuven repository. **Pinpoint:** PDF/printed pp. 11–14, 17–20, quantizer design, adopted training practices and TQT Table 2 results; bibliography p. 23, reference [12]. Source E-0081; marked copy E-1148.

Count: E-0459, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 12-Integer-Only-CNNs-MARKED.pdf

ARTICLE 16 · SUBSTANTIVE CITATION REPORT

A Reduced-Precision Network for Image Reconstruction

Manu Mathew Thomas; Karthik Vaidyanathan; Gabor Liktó; Angus G. Forbes
ACM Transactions on Graphics 39(6), Article 231 (2020) · 41 Google Scholar citations

Relies on or extends TQT

Documented contribution. University of California, Santa Cruz and Intel researchers train the quantization thresholds of their 4-bit QW-Net image-reconstruction network following Jain et al. [2019], training the thresholds in log space.

Direct methodological adoption. Explains that Jain et al. and Esser et al. showed quantization thresholds could be trained (p. 3), then states "Following Jain et al. [2019], we train the thresholds in log space" for QW-Net quantization (p. 6).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 11](#).

Reviewed version: ACM Transactions on Graphics publisher PDF. **Pinpoint:** Physical PDF pp. 3 and 6 (Section 5.1): trained quantization thresholds and log-space threshold training following Jain et al. [2019]; bibliography p. 11. Source E-1263; marked copy E-1399.

Count: E-0379, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 16-QW-Net-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 17 · SUBSTANTIVE CITATION REPORT

Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks

Sukru Burc Eryilmaz; Aysegul Dunder

IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737–10746 (2023) · 12 Google Scholar citations

Relies on or extends TQT

Documented contribution. NVIDIA and Bilkent University researchers use TQT as the baseline quantized retraining method for their study of orthogonality regularization, training thresholds in the log domain as described in the TQT paper and reporting results with and without TQT.

Direct methodological adoption. States "we use TQT [25] as our baseline retraining" (p. 5), trains thresholds in the log domain as described in [25] (p. 4) and combines the orthogonality penalty with TQT in Section V-A (p. 6).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 10](#).

Reviewed version: Final IEEE TNNLS print, vol. 34, no. 12; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF pp. 1–6 (printed pp. 10737–10742): TQT introduced as the quantized retraining method, log-domain threshold training as described in [25], TQT baseline retraining and experiments with TQT; bibliography p. 10 (printed p. 10746), reference [25]. Source E-1436; marked copy E-1400.

Count: E-0402, data.organic_results[19].inline_links.cited_by.total. Venue: E-1429.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 17-Orthogonality-MARKED.pdf

ARTICLE 18 · SUBSTANTIVE CITATION REPORT

FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration

Taha Soliman; Nellie Laleni; Tobias Kirchner; Franz Müller; Ashish Shrivastava; Thomas Kämpfe; Andre Guntoro; Norbert Wehn

ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022) · 50 Google Scholar citations

Relies on or extends TQT

Documented contribution. Robert Bosch GmbH, Fraunhofer IPMS and University of Kaiserslautern researchers evaluate their in-memory accelerator with 8-bit/4-bit quantization based on the optimizations presented in the TQT paper.

Direct methodological use. Section 6.2 states that the benchmark accuracy was tested using 8-bit/4-bit quantization based on the optimizations presented in [17], the TQT paper (p. 17).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 23](#).

Reviewed version: ACM Transactions on Embedded Computing Systems publisher PDF. **Pinpoint:** Physical PDF p. 17, Section 6.2: 8-bit/4-bit quantization based on the optimizations in [17]; bibliography p. 23, reference [17]. Source E-1279; marked copy E-1401.

Count: E-0381, data.organic_results[4].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 18-FELIX-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 19 · SUBSTANTIVE CITATION REPORT

7 μ J/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks

Georg Rutishauser; Moritz Scherer; Tim Fischer; Luca Benini

Future Generation Computer Systems 149 (2023), 717–731 · 12 Google Scholar citations

Relies on or extends TQT

Documented contribution. ETH Zürich and University of Bologna researchers train their ternarized gesture-recognition networks with two quantization-aware training algorithms, INQ and TQT, and report that ReLU activations with TQT give the highest classification accuracy.

Direct methodological use and comparison. Describes TQT as optimizing the logarithm of the clipping interval (p. 5), performs QAT with INQ and TQT (p. 6), lists TQT hyperparameters (p. 9) and compares TQT with INQ in Table 4 (p. 10).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 14](#).

Reviewed version: Published open-access journal PDF (Elsevier). **Pinpoint:** Physical PDF pp. 5–6, 9–10: TQT as one of two quantization-aware training algorithms, TQT hyperparameters and Table 4 TQT versus INQ results; bibliography p. 14, reference [67]. Source E-1254; marked copy E-1402.

Count: E-0382, `data.organic_results[9].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 19-DVS-Gesture-MARKED.pdf

ARTICLE 20 · SUBSTANTIVE CITATION REPORT

Optimizing the Deployment of Tiny Transformers on Low-Power MCUs

Victor J.B. Jung; Alessio Burrello; Moritz Scherer; Francesco Conti; Luca Benini

IEEE Transactions on Computers 74(2), 526–541 (2025) · 47 Google Scholar citations

Relies on or extends TQT

Documented contribution. Researchers in the ETH Zürich/University of Bologna PULP group quantize their tiny transformers with the uniform symmetric quantizer from the Trained Quantization Threshold method and define the TQT quantizer used in their deployment flow.

Direct methodological use. States "We use the uniform symmetric quantizer from the SotA Trained Quantization Threshold (TQT) method [3]" and gives the functional behavior of the TQT quantizer (p. 6).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 15](#).

Reviewed version: Published IEEE Transactions on Computers print; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 6 (printed p. 531), Section IV-C: the uniform symmetric TQT quantizer used in the QuantLib flow; bibliography p. 15 (printed p. 540), reference [3]. Source E-1434; marked copy E-1403.

Count: E-0375, `data.organic_results[6].inline_links.cited_by.total`. Venue: E-1421.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 20-Tiny-Transformers-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 21 · SUBSTANTIVE CITATION REPORT

A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps

Xiaowen Jiang; Valerio Camballeri; Gianluca Agresti; Cynthia Ifeyinwa Ugwu; Adriano Simonetto; Fabien Cardinaux; Pietro Zanuttigh

CVPR Workshops 2022 (CVPRW), 2686–2695 · 12 Google Scholar citations

Relies on or extends TQT

Documented contribution. EPFL, Sony and University of Padova researchers report that their most effective uniform-precision quantization-aware training procedure initializes from the pretrained model and uses learnable qmax as in the Trained Quantization Thresholds approach.

Direct methodological use. States that the most effective uniform-precision QAT procedure used learnable qmax as in the TQT approach [23] (p. 5).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 9.](#)

Reviewed version: IEEE Xplore published print, CVPRW 2022 proceedings; public excerpt reproduces the title, discussion and bibliography pages. **Pinpoint:** Physical PDF p. 5, Section 5.1 (printed p. 2690): uniform-precision QAT using learnable qmax as in TQT [23]; bibliography p. 9 (printed p. 2694), reference [23]. Source E-1435; marked copy E-1404.

Count: E-0402, data.organic_results[14].inline_links.cited_by.total. Venue: E-1422.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 21-Depth-Completion-MARKED.pdf

ARTICLE 22 · SUBSTANTIVE CITATION REPORT

A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition

Jaemyung Kim; Jin-Ku Kang; Yongwoo Kim

IEEE Access 9 (2021), 104367–104381 · 54 Google Scholar citations

Relies on or extends TQT

Documented contribution. Inha University and Sangmyung University researchers develop log level threshold quantization (LLTQ) as a hardware-friendly alternative to TQT's power-of-two scale mapping and compare LSQ, TQT and LLTQ in quantization-aware training experiments.

Method extension and implemented comparison. Describes TQT as mapping the scale factor to power-of-two terms through a log-threshold (p. 6), proposes LLTQ to avoid TQT's pre-calibration step, illustrates LSQ, TQT and LLTQ in Figure 3 (p. 7) and reports that TQT is hardware-friendly but loses about 0.45% accuracy versus floating point, where LLTQ does not (p. 11).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 15.](#)

Reviewed version: IEEE Access open-access publisher PDF. **Pinpoint:** Physical PDF pp. 3, 6–7, 10–11: TQT in related work, LLTQ derived as an alternative to TQT's power-of-two scale mapping, Figure 3 and the implemented LSQ/TQT/LLTQ comparison; bibliography p. 15, reference [39]. Source E-1264; marked copy E-1405.

Count: E-0393, data.organic_results[0].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 22-LLTQ-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 23 · SUBSTANTIVE CITATION REPORT

Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach

Rubén Padial-Allué; Alberto Martín-Martín; Antonio García; Encarnación Castillo; José David Fernández-Rodríguez; Marcos Baena-Molina; Luis Parrilla

Computers and Electrical Engineering 134 (2026), 111101 · No cited-by count in the saved Google Scholar record

Relies on or extends TQT

Documented contribution. University of Granada-led researchers use a power-of-two scale quantizer with TQT in the quantization-aware training for their AMD DPU implementation. Other activation and training changes accompany TQT, so results are not an isolated TQT effect.

Direct methodological use. States "For the quantization strategy, a power-of-two scale quantizer with training quantization threshold (TQT) was used [50]" in the AMD DPU quantization-aware training (p. 9).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 17](#).

Reviewed version: Open-access Elsevier PDF (CC BY), University of Granada repository copy. **Pinpoint:** Physical PDF p. 9, Section 3.2: power-of-two scale quantizer with TQT for the AMD DPU implementation; bibliography p. 17, reference [50]. Source E-1185; marked copy E-1406.

Count: E-0385, data.organic_results[1].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 23-RIS-MARKED.pdf

ARTICLE 24 · SUBSTANTIVE CITATION REPORT

INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization

Ninnart Fuengfusin; Hakaru Tamukoh

Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171–176 · No cited-by count in the saved Google Scholar record

Relies on or extends TQT

Documented contribution. Kyushu Institute of Technology researchers build ITBWN by quantizing activations to INT8 with TQT while using binary or ternary weights. They use the Xilinx Brevitas TQT implementation and note that its settings omit the power-of-two scaling constraint.

Direct methodological use. States that ITBWN uses TQT INT8 quantization for its activations (p. 3) and labels its binary, ternary and INT8 configurations by their use of TQT (p. 4).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 5](#).

Reviewed version: Published journal PDF (J-STAGE). **Pinpoint:** Physical PDF pp. 2–5: TQT as the INT8 method, activations quantized with TQT via Xilinx Brevitas, TQT-based settings and conclusion; bibliography p. 5, reference 11. Source E-0595; marked copy E-1407.

Count: E-0422, data.organic_results[9].inline_links.cited_by.total.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 24-ITBWN-MARKED.pdf

B. ARTICLES RELYING ON OR EXTENDING TQT

ARTICLE 25 · SUBSTANTIVE CITATION REPORT

FPGA-Based Real-Time Embedded Fish Embryo Detection System

Mengqi Wang; Guofu Feng; Ming Chen; Ruijuan Ye; Yaohui Wang

Mobile Information Systems, vol. 2022, 8021288 · No cited-by count in the saved Google Scholar record

Relies on or extends TQT

Documented contribution. Shanghai Ocean University and Nantong Long Yang Aquatic Company researchers use the trained quantization threshold method to quantize their pruned YOLOv3 detector to 4-bit fixed point for FPGA deployment. Reference [33] prints the author list in garbled order.

Direct methodological use. States that the TQT method is used to transform the pruned FP32 model into a 4-bit fixed-point model during training (p. 4); the quantized model is deployed on the FPGA.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 15](#).

Reviewed version: Open-access Hindawi/Wiley publisher PDF (CC BY). **Pinpoint:** Physical PDF p. 4, Section 3.2.2: TQT used to quantize the pruned YOLOv3 model to INT4; bibliography p. 15, reference [33]. Source E-1281; marked copy E-1408.

Count: E-0398, `data.organic_results[0].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 25-Fish-Embryo-MARKED.pdf

ARTICLE 26 · SUBSTANTIVE CITATION REPORT

Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing

Colin Laganier; Liam Fletcher; Elim Kwan; Richard Walters; Victoria Nockles

SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025) · 6 Google Scholar citations

Relies on or extends TQT

Documented contribution. The Alan Turing Institute researchers deploy a YOLOv8-based SAR vessel detector on an AMD/Xilinx Kria KV260 FPGA. They quantize it to INT8 with the Vitis AI Quantizer using power-of-two scaling and quantization-aware training attributed to TQT, train the quantizer parameters and deploy the resulting QAT models.

Methodological use via Vitis AI. Cites TQT [35] for its power-of-two QAT with a straight-through estimator (p. 5), retrains with a tuned quantizer parameter learning rate and reports the deployed FPGA QAT models in Table 4 (p. 11). TQT is cited rather than named, and was used through the AMD/Xilinx Vitis AI toolchain.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. [Bibliography p. 16](#).

Reviewed version: Published ACM version (open access, CC BY-NC 4.0). **Pinpoint:** Physical PDF p. 5, Section 3.2: INT8 Vitis AI quantization with power-of-two scaling and QAT using a straight-through estimator [35]; p. 11, Section 4.2.2, Table 4 and Section 4.3: QAT with a quantizer parameter learning rate and the deployed FPGA QAT models; bibliography p. 16, reference [35]. Source E-1277; marked copy E-1438.

Count: E-0386, `data.organic_results[11].inline_links.cited_by.total`.

[Original article](#) · [Scholar cited-by list](#)

Marked file: 26-SAR-Vessel-MARKED.pdf

Citation coverage across the research portfolio

Entry	Cited work	Profile citations	Records retrieved	Grouped works	Mapped works
01	Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks	227	216	206	146
02	Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware	71	71	69	41
03	Degradation of 2DEG transport properties in GaN-capped AlGaIn/GaN heterostructures at 600° C in oxidizing and inert environments	22	22	20	11
04	Interdigitated Pt-GaN Schottky interfaces for high-temperature soot-particulate sensing	15	15	15	5
05	Training a fully convolutional neural network to route integrated Circuits	13	13	13	6
06	A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections (Part 2)	4	4	4	3
07	System and method for implementing neural networks in integrated circuits	1	1	1	0
08	Gallium Nitride Microelectronics for High-Temperature Environments	1	1	1	1
09	An ML Compilation Stack for Autonomous Vehicles using MLIR	0	0	0	0
10	Systematic DL Optimizations for Accurate and Efficient TensorRT Deployments on Cruise's Self-Driving Fleet	0	0	0	0
11	TQT: Trained Quantization Thresholds	0	0	0	0

A citing work can appear in several publication rows. Unique survey totals group overlapping records and publication versions.

The saved Scholar profile records 354 citations. The inventory retains 323 works from 337 archived Scholar records. Publication affiliations establish locations for 210 works; the remaining 113 have their review status recorded in the inventory.

Dated Scholar responses document discovery and article-level counts. Source PDFs establish the selected discussions and affiliation evidence. The coverage table and inventory provide the denominators for each measure.

Publication-sourced affiliations and research connections

The map presents 228 institution/location entries from 210 grouped citing works in 30 countries/territories. Each pin opens the institutions, citing papers and affiliation-page evidence. Several institutions in one city share a display pin; multi-institution papers contribute several locations.

Affiliations come from the reviewed publication. Explicit campus and city details determine placement; ROR supplies institution-city geography when the publication lists an institution without a city. Hollow markers show country-level affiliations. Coordinates are geographic display anchors.

The map documents the geographic reach of citing authors' publication affiliations, including coauthor- and employer-linked works. Article passages establish methodological adaptation, implementation, comparison, design connections or technical exposition. Joint attribution is retained where several methods are credited.

Research files

[Marked discussions](#): pages 1–103 in this report's article order.

Marked-Articles/: complete articles with original page numbers and source links.

Selected-Articles.csv: selection membership, counts, evidence types and source pages.

Discussion-Page-Index.csv: article-to-packet page mapping.

Sources/: dated Scholar, publication, venue, geography and affiliation captures.

[Substantive Citation Report](#) · [Interactive map](#)

Counts refer to the saved October 7, 2026 snapshot. Hashes and source provenance are recorded in the research manifests.